Support-Controlled Harmonic Reversal Asymmetry and Exact Inversion-Quotient Allocation in Bach Chorales and WJazzD

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Abstract

We estimate forward-versus-reversed held-out log scores for transposition-quotiented harmonic transitions in 430 Bach chorales (music21 corpus, 26249 chord changes, 67 key-relative states) and 412 Weimar Jazz Database solos grouped into 273 compositions (27491 chord changes, 175 states). At bilateral training support $r = 1$, scores across $\alpha \in \{0.1, 0.5, 1\}$ range from 0.765 to 0.832 bits per transition in Bach and from 1.555 to 2.670 in WJazzD, depending on retained-total versus conditional scoring. A standard KL chain-rule identity gives both a nonnegative population decomposition and an exact eventwise decomposition of held-out log scores under an inversion quotient. The observed inversion-resolving held-out term is 0.137 to 0.156 bits in Bach, 16.0 to 16.2% of its score, with equal-chorale 95% lower bounds of 0.128 to 0.145. WJazzD yields 0.091 to 0.151 bits, or 3.7 to 4.7%, with composition-level lower bounds of 0.053 to 0.089, below the pre-specified 0.10-bit threshold. Three controls qualify the share contrast: shared-support vocabulary standardisation attenuates 30.5% of the $\alpha = 0.5$ gap but covers only three shared strata; fold-local alphabet matching at $k \in \{25, 50, 67\}$ leaves both shares in place; and a mode split shows that the Bach term is a minor-mode quantity (0.1% of the major-mode score, 29.9% of the minor-mode score), so that common-mode weighting gives 11.5% against 4.8%. Stationary circulation decompositions close exactly and are more diffuse in WJazzD; named-window counts show that a high cycle affinity can rest on a single occurrence. This extends González-Espinoza, Martínez-Mekler and Lacasa (2020) from melodic visibility statistics to harmonic transition laws; no theorem novelty is claimed.

I. Logic

1 Prior musical irreversibility and claim boundary

Time irreversibility of music has been quantified before: González-Espinoza, Martínez-Mekler and Lacasa (2020) applied horizontal-visibility-graph statistics to scalar melodic note streams across 8,856 pieces. We quantify a different object, the reversal asymmetry of transposition-quotiented harmonic transition laws, and decompose it through inversion-sensitive, directed-edge and closed-cycle analyses. The quotient identity we use is a corollary of the KL chain rule and of exact coarse-graining results (Esposito, 2012; Teza and Stella, 2020); it is stated as a lemma because it fixes the estimand and prevents an incorrect quotient implementation, not as a contribution.

2 Harmonic edge laws, reversal and bilateral-support estimands

Definition 1. *A chord event is a key-relative 12-bit pitch-class mask with mode (the annotated tonic at pitch class 0); consecutive identical masks are collapsed to chord changes (an uncollapsed* *control is reported). A directed edge is a pair (i,j) of consecutive states; $R(i,j)=(j,i)$ is reversal. For an edge law μ, $\sigma(\mu)=D_{\mathrm{KL},2}(\mu\|R_*\mu)$ in bits.*

Held-out scoring: for work-grouped folds, each held-out edge receives $d_{ij}=\log_2\frac{c_{ij}+\alpha}{c_{ji}+\alpha}$ from training counts; the event mean $\bar{s}$ is a predictive log score, not σ of a law. Bilateral support restricts to edges with $c_{ij},c_{ji}\ge r$ in training; the retained-total estimand assigns excluded edges zero, the conditional estimand averages over retained edges only. Corpus scores weight held-out events equally. Group inference first averages the score within each chorale or composition and then weights groups equally; the one-sided normal 95% LCB is $\bar{d}_G-1.645\,s_G/\sqrt{G}$ (Tables 3 and 4; Table 2 uses 1.96 as computed under the pre-specified rule). Because the weightings differ, a group LCB need not lie below the event-weighted corpus score.

3 Diagonal C_{12} quotient

Let $G=\mathbb{Z}_{12}$ act diagonally on edges, $T_g(i,j)=(gi,gj)$, and let reversal commute with the action. For the symmetrised law $\mu^G=\frac{1}{12}\sum_g(T_g)_*\mu$ and the orbit map q, $\sigma(\mu^G)=\sigma(q_*\mu)$ (KL chain rule for the statistic q; both symmetrised laws are uniform within each orbit, so the conditional term vanishes). Our key-relative representation is the tonic-anchored orbit representative; an exact unsmoothed check on the common support gives identity gap 0.0 in Bach. Smoothed plug-ins differ because pseudocounts must be pushed forward, not added per cell.

4 Exact $C_{12}\to D_{12}$ allocation

Lemma 1 (standard KL chain-rule corollary)**.** *Let h be a quotient commuting with reversal, P an edge law and $Q=R_*P$. Then*

$$D_{\mathrm{KL}}(P\|Q)=D_{\mathrm{KL}}(h_*P\|h_*Q)+\sum_z(h_*P)(z)\,D_{\mathrm{KL}}\big(P(\cdot\mid z)\|Q(\cdot\mid z)\big).$$

The second population term is nonnegative. For a fold-fitted predictive pair $(\widehat{P}_f,\widehat{Q}_f)$ the corresponding held-out identity is eventwise,

$$\log_2\frac{\widehat{P}_f(x)}{\widehat{Q}_f(x)}=\log_2\frac{h_*\widehat{P}_f(hx)}{h_*\widehat{Q}_f(hx)}+\log_2\frac{\widehat{P}_f(x\mid hx)}{\widehat{Q}_f(x\mid hx)},$$

and its held-out residual is not guaranteed nonnegative by the theorem; positivity and its group LCB are empirical results.

Here κ is mode-conditioned pitch-class inversion ($p\mapsto -p$, mode fixed) and h identifies an edge with its inversion; $h_*\widehat{P}_f$ is obtained only by pushforward. We write $\bar{s}_{C_{12}}=\bar{s}_{D_{12}}+\bar{\delta}_{\mathrm{inv}}$ for the held-out event means. $\bar{\delta}_{\mathrm{inv}}$ is the part of the reversal score that requires distinguishing an interval structure from its inversion; it can differ from zero only on events whose inverted edge has training mass. The mode-fixed action is one valid D_{12} action, not the unique music-theoretic one.

5 Detailed balance, cycle affinities and circulation

For a stationary first-order chain with flow $F_{ij}=\pi_iP_{ij}$, $\sigma=\sum_{i<j}(F_{ij}-F_{ji})\log_2(F_{ij}/F_{ji})$, and $\sigma=0$ iff detailed balance iff every cycle affinity $A_C=\sum_{(i,j)\in C}\log_2(F_{ij}/F_{ji})$ vanishes (Kolmogorov,

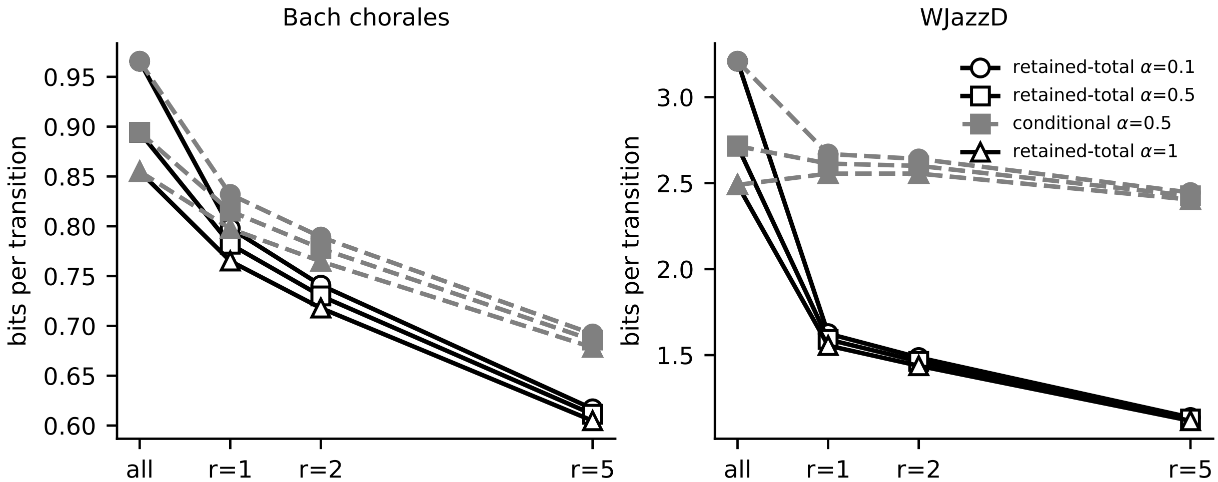

Figure 1: Bilateral-support control: retained-total (solid) and conditional (dashed) scores across support thresholds and smoothing.

1936). With a spanning forest, $\sigma = \sum_C j_C A_C$ over fundamental cycles; individual terms are basis dependent, the total is not. A loop's affinity is a force, not a contribution: it is reported together with its circulation and with contiguous window counts.

II. Experiment

6 Corpora, states, grouped folds and pre-specification

Bach: 430 chorales, 26249 collapsed chord changes, 67 states; folds by chorale. WJazzD: 412 solos in 273 compositions, 27491 collapsed chord changes, 175 states; folds by composition so performances of one tune never cross folds. Chord types are triads (Bach, music21 quality) and thirteen jazz symbol templates (WJazzD). State alphabets differ (67 versus 175); raw magnitudes are not compared across corpora. The conjunctive decision rule (equal-group 95% LCB of $\bar{\delta}_{\text{inv}}$ above 0.10 bits in both corpora at every $\alpha \in \{0.1, 0.5, 1\}$, closure error below 10^{-12}) was fixed in the dated analysis plan of 2 September 2026 before the inversion allocation was computed; it was recorded in the project ledger, not deposited in a public registry, so we call it pre-specified rather than preregistered. The vocabulary, alphabet, mode and cycle analyses are exploratory.

7 Support-controlled held-out irreversibility

Endpoint (piece boundary) marginals contribute 0.2% (Bach) and 0.0% (WJazzD) of the plug-in σ; merging states with fewer than five occurrences changes nothing. Figure 1 shows the bilateral-support curves.

Table 1: Held-out reversal asymmetry by bilateral support (bits per transition). rho = share of held-out events on edges observed in both directions in training; retained-total assigns excluded edges zero; LCB = work- (Bach) or composition- (WJazzD) level 95% lower bound.

α	support	ρ (%)	retained-total	conditional	LCB	works with no retained edge
Bach chorales						
0.1	all	100.0	0.966	0.966	0.975	0
0.1	r=1	95.9	0.798	0.832	0.809	0
0.1	r=2	93.9	0.741	0.789	0.747	0
0.1	r=5	89.1	0.617	0.692	0.618	0
0.5	all	100.0	0.894	0.894	0.904	0
0.5	r=1	95.9	0.782	0.815	0.792	0
0.5	r=2	93.9	0.730	0.778	0.736	0
0.5	r=5	89.1	0.611	0.686	0.613	0
1	all	100.0	0.855	0.855	0.865	0
1	r=1	95.9	0.765	0.798	0.775	0
1	r=2	93.9	0.718	0.765	0.724	0
1	r=5	89.1	0.605	0.678	0.606	0
WJazzD						
0.1	all	100.0	3.209	3.209	2.700	0
0.1	r=1	60.9	1.625	2.670	1.411	14
0.1	r=2	56.3	1.486	2.641	1.297	17
0.1	r=5	46.5	1.137	2.445	0.962	20
0.5	all	100.0	2.715	2.715	2.304	0
0.5	r=1	60.9	1.591	2.614	1.382	14
0.5	r=2	56.3	1.463	2.600	1.276	17
0.5	r=5	46.5	1.129	2.426	0.954	20
1	all	100.0	2.490	2.490	2.120	0
1	r=1	60.9	1.555	2.555	1.350	14
1	r=2	56.3	1.438	2.556	1.254	17
1	r=5	46.5	1.118	2.404	0.945	20

8 Inversion allocation and group inference

The pre-specified conjunctive rule fails: the minimum LCB is 0.053 (WJazzD, $\alpha = 1$). Bach passes at every α and its share is stable at 16.0 to 16.2% across collapsed and uncollapsed events; WJazzD's share is 3.7 to 4.7%. The absolute residuals overlap (0.137 to 0.156 versus 0.091 to 0.151 bits); the share contrast reflects WJazzD's roughly three-fold larger inversion-invariant score.

9 Controls on the share contrast: vocabulary, alphabet and mode

At $\alpha = 0.5$, standardising to the 3 inversion-closed chord-family strata shared by both corpora changes the inversion shares from 16.0% and 4.0% to 15.9% and 7.6%, reducing the gap from 0.120 to 0.083 ($E = 0.30$). This restricted shared-support standardisation therefore does not remove most of the observed contrast, but it excludes the 29 jazz-only strata (two of which hold 48% of WJazzD events) and does not establish a vocabulary-independent corpus effect.

Matching alphabets at $k \in \{25, 50, 67\}$ leaves the WJazzD share at 4.2 to 4.8% (LCB 0.082 to 0.100) and the Bach share at 16.0 to 25.4% at $\alpha = 0.5$ (the Bach share rises at $k = 25$ because the retained events concentrate on the most frequent, inversion-paired triads); WJazzD retains 75.0% of test mass at $k = 67$ against 99.7% for Bach. The share difference is therefore not removed by

Table 2: Exact eventwise allocation of held-out reversal log scores (bits per transition). Per-event closure error $\leq 9 \times 10^{-16}$. Group = chorale (Bach) or composition (WJazzD); LCB at 1.96 as pre-specified.

corpus	α	$\bar{s}_{C_{12}}$	$\bar{s}_{D_{12}}$	$\bar{\delta}_{\text{inv}}$	share (%)	group mean	group LCB
Bach	0.1	0.966	0.809	0.156	16.2	0.166	0.145
Bach	0.5	0.894	0.752	0.143	16.0	0.152	0.133
Bach	1	0.855	0.719	0.137	16.0	0.146	0.128
WJazzD	0.1	3.209	3.058	0.151	4.7	0.120	0.089
WJazzD	0.5	2.715	2.607	0.108	4.0	0.085	0.063
WJazzD	1	2.490	2.399	0.091	3.7	0.071	0.053
Bach (uncollapsed)	0.5	0.667	0.561	0.106	16.0	0.117	0.102
WJazzD (uncollapsed)	0.5	2.694	2.587	0.108	4.0	0.084	0.063

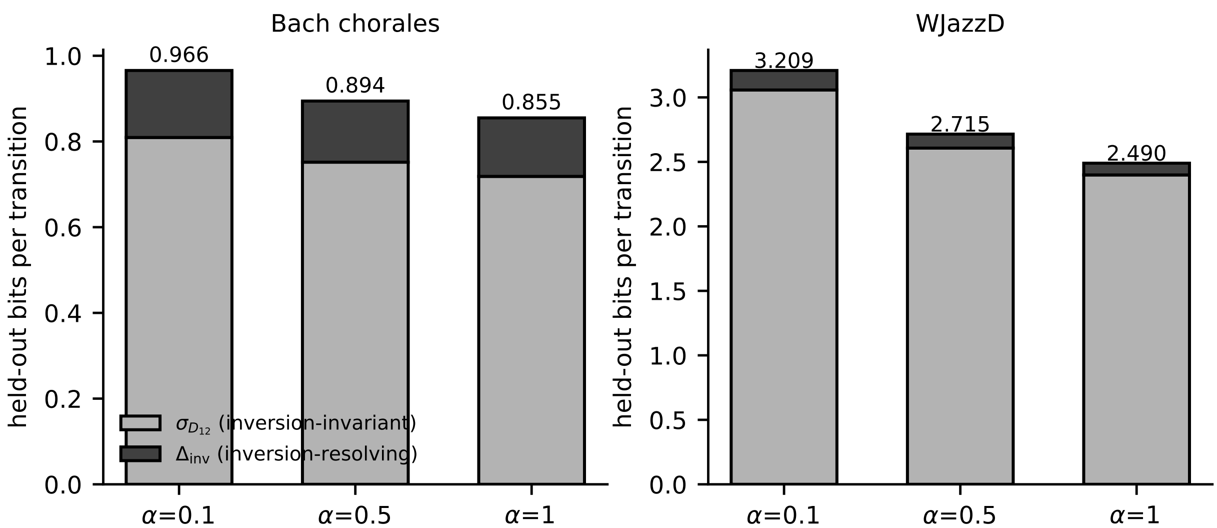

Figure 2: Exact allocation of held-out reversal scores into inversion-invariant and inversion-resolving parts.

alphabet size.

The Bach inversion-resolving share is a minor-mode quantity: 0.1% of the major-mode score against 29.9% of the minor-mode score, with 43.9% of Bach events in minor against 16.2% in WJazzD (WJazzD major 3.1%, minor 9.9%). Under common-mode weights the shares are 11.5% (Bach) and 4.8% (WJazzD): the mode mixture accounts for part of the raw contrast, and within the major mode the ordering reverses. The κ-support column shows the mechanism: in Bach major only 6.8% of held-out edges have an inverted edge with training mass, so the inversion quotient leaves almost every score unchanged, whereas in Bach minor 84.0% do. Taken together: the raw 16% versus 4% contrast is not removed by matched alphabets or by restricted shared-support standardisation, but it is representation-dependent through mode. The supported statement is that inversion orientation carries a reversal-relevant share of the score in Bach minor-mode progressions and a small share elsewhere; no genre effect is identified.

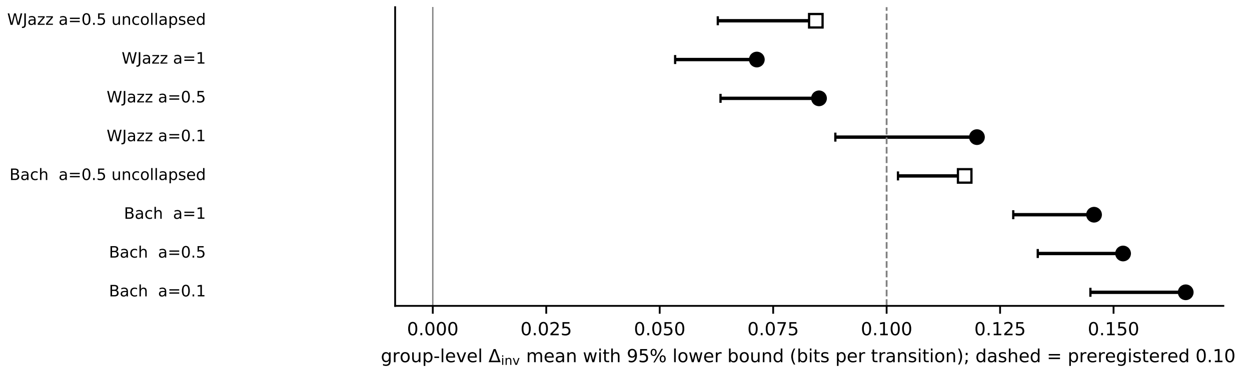

Figure 3: Group-level $\bar{\delta}_{\text{inv}}$ with 95% lower bounds; dashed = pre-specified 0.10 bits.

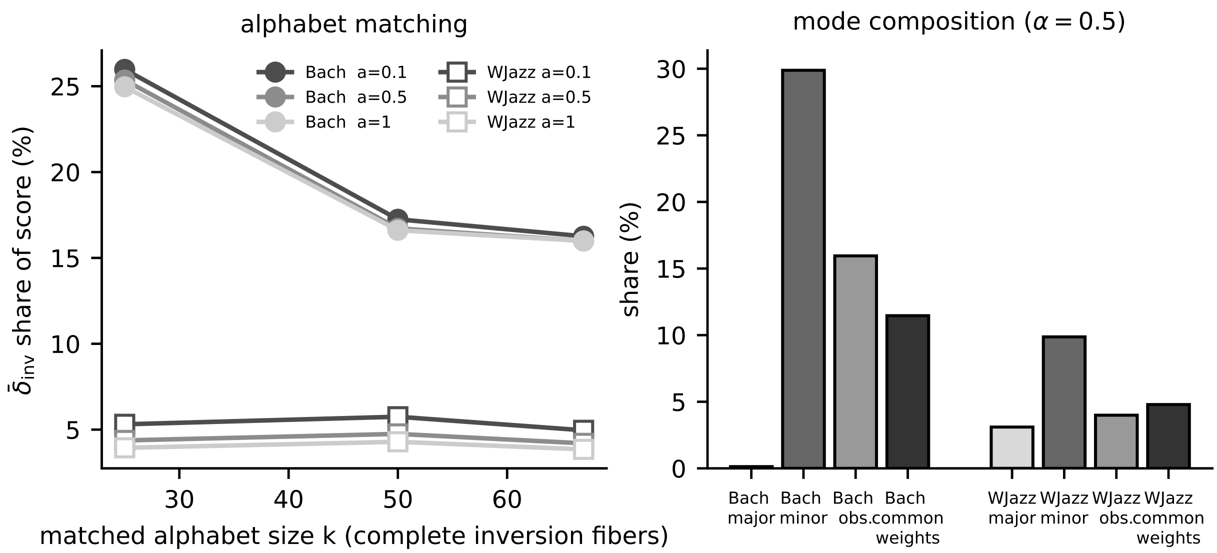

Figure 4: Left: inversion-resolving share under fold-local alphabet matching. Right: share by held-out mode, observed mixture and common-mode weighting ($\alpha = 0.5$).

10 Directed edges, named windows and stationary cycle circulation

The exploratory secondary-dominant cycle $II^7 \to V^7 \to I^{maj7}$ has the largest edge-product affinity in WJazzD but is supported by one complete forward window in one composition, whereas the conventional ii^7 cycle occurs 54 times in 17 compositions and is never observed in reverse (Table 6). In this spanning-forest basis Bach's stationary circulation is more concentrated than WJazzD's among the five largest terms; these percentages are basis dependent and do not mean that five named progressions explain those fractions of corpus-level asymmetry.

Table 3: Alphabet matching. In each training fold the state alphabet is restricted to the highest-mass complete inversion fibers up to target size k (a fiber is never split; test counts are not used); edges with an endpoint outside the alphabet are dropped from training and test. Achieved size, fixed (self-inverse) and paired fibers, retained test-event mass, groups with no retained event, absolute inversion-resolving score $\bar{\delta}_{\mathrm{inv}}$, its share of $\bar{s}_{C_{12}}$, and the one-sided equal-group 95% LCB.

corpus	α	target k	achieved	fixed	paired	retained mass (%)	empty groups	$\bar{\delta}_{\mathrm{inv}}$	share (%)	LCB
Bach	0.1	25	26	0	13	64.6	0	0.162	26.0	0.152
Bach	0.1	50	49	1	24	95.8	0	0.165	17.2	0.155
Bach	0.1	67	68	6	31	99.7	0	0.157	16.3	0.149
WJazzD	0.1	25	24	0	12	46.3	22	0.163	5.3	0.130
WJazzD	0.1	50	53	1	26	67.2	10	0.199	5.8	0.138
WJazzD	0.1	67	74	2	36	75.0	7	0.177	5.0	0.113
Bach	0.5	25	26	0	13	64.6	0	0.154	25.4	0.145
Bach	0.5	50	49	1	24	95.8	0	0.150	16.7	0.142
Bach	0.5	67	68	6	31	99.7	0	0.143	16.0	0.137
WJazzD	0.5	25	24	0	12	46.3	22	0.128	4.4	0.097
WJazzD	0.5	50	53	1	26	67.2	10	0.147	4.8	0.100
WJazzD	0.5	67	74	2	36	75.0	7	0.131	4.2	0.082
Bach	1	25	26	0	13	64.6	0	0.149	25.0	0.140
Bach	1	50	49	1	24	95.8	0	0.143	16.6	0.136
Bach	1	67	68	6	31	99.7	0	0.137	16.0	0.131
WJazzD	1	25	24	0	12	46.3	22	0.111	3.9	0.083
WJazzD	1	50	53	1	26	67.2	10	0.125	4.3	0.084
WJazzD	1	67	74	2	36	75.0	7	0.112	3.9	0.069

11 Limitations and replication

These results describe two annotated corpora under fixed event definitions, chord templates, key estimates (machine-estimated for Bach, annotated for WJazzD), smoothing rules, and a mode-conditioned inversion that leaves the binary mode covariate fixed. The held-out quantities are predictive log scores, not physical entropy-production estimates. Bach and WJazzD have unequal state alphabets (67 versus 175) and no genre effect is identified. The vocabulary standardisation covers only three shared strata and excludes 29 jazz-only strata. The stationary cycle analysis uses $\alpha = 0.5$, which gives positive fitted flow to pseudocount-supported edges; individual fundamental cycles, ranks and shares depend on the spanning forest and smoothing choice. Named-window counts establish corpus occurrence, not functional or causal harmonic syntax. Code and derived counts are released; every table regenerates from the JSON manifest.

References

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Table 4: Mode composition. Training uses all events; held-out events are split by the annotated mode of the source chord. κ-support = share of held-out edges whose inverted edge has positive training count (the only events on which the inversion quotient can change a score). Common-mode weights = pooled major/minor event shares of both corpora (70.2% major at $\alpha = 0.5$). LCB = one-sided equal-group 95%.

corpus	α	held-out events	n	κ-support (%)	$\bar{s}_{C_{12}}$	$\bar{\delta}_{\text{inv}}$	share (%)	LCB
Bach	0.1	major only	14718	6.8	0.795	0.001	0.1	-0.000
Bach	0.1	minor only	11531	84.0	1.183	0.355	30.0	0.335
Bach	0.1	observed mixture	26249	40.7	0.966	0.156	16.2	0.148
Bach	0.1	common-mode weights	–	–	0.911	0.106	11.7	–
WJazzD	0.1	major only	23031	8.3	3.268	0.115	3.5	0.079
WJazzD	0.1	minor only	4460	10.2	2.903	0.336	11.6	0.113
WJazzD	0.1	observed mixture	27491	8.6	3.209	0.151	4.7	0.094
WJazzD	0.1	common-mode weights	–	–	3.159	0.181	5.7	–
Bach	0.5	major only	14718	6.8	0.747	0.001	0.1	0.000
Bach	0.5	minor only	11531	84.0	1.082	0.324	29.9	0.308
Bach	0.5	observed mixture	26249	40.7	0.894	0.143	16.0	0.136
Bach	0.5	common-mode weights	–	–	0.847	0.097	11.5	–
WJazzD	0.5	major only	23031	8.3	2.814	0.087	3.1	0.059
WJazzD	0.5	minor only	4460	10.2	2.206	0.218	9.9	0.071
WJazzD	0.5	observed mixture	27491	8.6	2.715	0.108	4.0	0.067
WJazzD	0.5	common-mode weights	–	–	2.633	0.126	4.8	–
Bach	1	major only	14718	6.8	0.721	0.001	0.1	0.000
Bach	1	minor only	11531	84.0	1.026	0.309	30.1	0.295
Bach	1	observed mixture	26249	40.7	0.855	0.137	16.0	0.131
Bach	1	common-mode weights	–	–	0.812	0.093	11.4	–
WJazzD	1	major only	23031	8.3	2.602	0.076	2.9	0.051
WJazzD	1	minor only	4460	10.2	1.909	0.172	9.0	0.054
WJazzD	1	observed mixture	27491	8.6	2.490	0.091	3.7	0.056
WJazzD	1	common-mode weights	–	–	2.396	0.104	4.3	–

Table 5: Top directed edges by plug-in contribution to σ (forward and reverse counts).

edge	forward	reverse	bits
Bach			
V maj (major) -> I maj (major)	1611	839	0.058
II dim (minor) -> V maj (minor)	193	4	0.040
V maj (minor) -> I maj (minor)	167	2	0.039
IV maj (major) -> V maj (major)	429	87	0.037
IV min (minor) -> V maj (minor)	214	10	0.035
VII dim (major) -> I maj (major)	390	76	0.035
II maj (major) -> V maj (major)	463	117	0.035
bVII maj (minor) -> bIII maj (minor)	579	206	0.033
WJazzD			
II m7 (major) -> V 7 (major)	1976	107	0.302
V 7 (major) -> I maj7 (major)	984	42	0.162
VI 7 (major) -> II m7 (major)	975	54	0.148
III m7 (major) -> VI 7 (major)	431	5	0.099
V 7 (major) -> I 6 (major)	438	24	0.066
VI 7 (major) -> II 7 (major)	279	3	0.064
I 7 (major) -> IV maj7 (major)	196	0	0.061
II 7 (major) -> V 7 (major)	332	12	0.057

Table 6: Panel A. Named progressions as contiguous windows within works (collapsed events). $N^+ : N^-$ = complete forward and exactly reversed occurrences; the smoothed window ratio is $\log_2 \frac{N^+ + 0.5}{N^- + 0.5}$, not the edge-product affinity A_C. An open path is not a cycle and has no affinity.

corpus	progression	status	$N^+ : N^-$	works with forward	window ratio (bits)
Bach	$i \to iv \to V \to i$	closed	20:0	19	5.358
Bach	$I \to IV \to V \to I$	closed	115:4	94	4.682
Bach	$I \to ii \to V \to I$	closed	165:8	109	4.283
WJazzD	$Imaj7 \to ii7 \to V7 \to Imaj7$	closed	54:0	17	6.768
WJazzD	$Imaj7 \to II7 \to V7 \to Imaj7$	closed	1:0	1	1.585
WJazzD	$Imaj7 \to vi7 \to ii7 \to V7$	open path	0:0	0	0.000

Table 7: Panel B. Spanning-forest circulation terms of the stationary first-order fit ($\alpha = 0.5$; maximum symmetric-traffic forest; return state printed). $\sigma = \sum_C j_C A_C$ closes with error 0.0e+00 (Bach, $\sigma = 0.864$, 2,145 cycles) and 0.0e+00 (WJazzD, $\sigma = 2.156$, 15,051 cycles). Top-5 shares 28.2% and 15.6%, top-10 shares 45.4% and 25.8%. Terms, ranks and shares are basis dependent; only the total is invariant. Stationary σ is a model quantity and is not comparable with the held-out scores of Table 1.

corpus	fundamental cycle	j_C	A_C (bits)	$j_C A_C$	share (
Bach	$Vmajm \to IIdimm \to Iminm \to Vmajm \to Vmajm$	-0.00642	-8.51	0.0546	
Bach	$Vmaj \to IVmaj \to Imaj \to Vmaj \to Vmaj$	-0.01214	-4.37	0.0530	
Bach	$Vmajm \to IVminm \to bIIImajm \to Iminm \to Vmajm \to Vmajm$	-0.00735	-6.49	0.0477	
Bach	$VIIdim \to IVmaj \to Imaj \to VIIdim \to VIIdim$	-0.00642	-7.33	0.0471	
Bach	$Vmaj \to IImin \to Imaj \to Vmaj \to Vmaj$	-0.01098	-3.75	0.0412	
WJazzD	$IIm7 \to I6 \to V7 \to IIm7 \to IIm7$	-0.00604	-13.50	0.0815	
WJazzD	$II7 \to I6 \to V7 \to II7 \to II7$	-0.00576	-13.37	0.0770	
WJazzD	$Imaj7 \to VI7 \to IIm7 \to V7 \to Imaj7 \to Imaj7$	0.00400	16.20	0.0647	
WJazzD	$I7 \to IIIm7 \to VI7 \to IIm7 \to V7 \to I7 \to I7$	0.00249	24.08	0.0600	
WJazzD	$Imaj \to VI7 \to IIm7 \to V7 \to Imaj \to Imaj$	0.00283	18.69	0.0530	

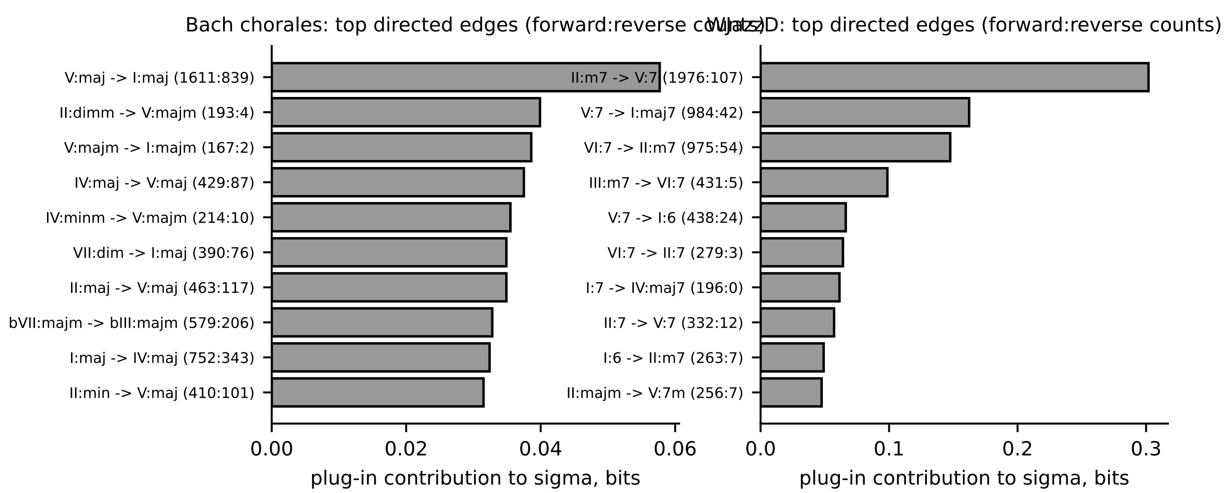

Figure 5: Top directed edges by contribution.

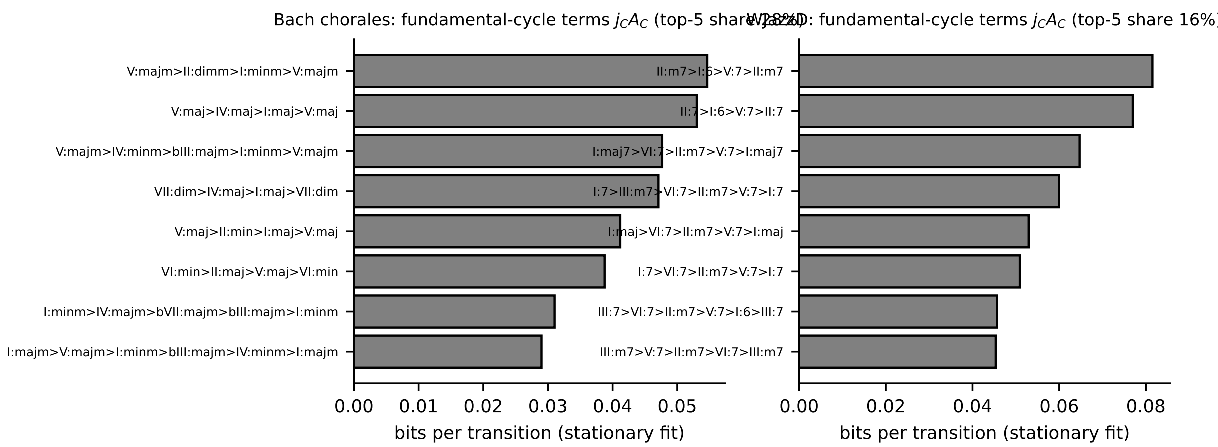

Figure 6: Fundamental-cycle terms of the stationary fit.